

INTERNATIONAL UNIVERSITY WESTERN CAMPUS
 FACULTY OF BIOMEDICAL SCIENCES
 DEPARTMENT OF MICROBIOLOGY AND IMMUNOLOGY
 Supplementary examinations march 2010
 Basic Immunology 1
 Duration: 2 hours 30 mins
 Class: BMS 1.2

Reg No.

Instructions:

Attempt all questions in Section A and FIVE (5) questions from Section B.

SECTION A (40 marks) - answer all questions

1. Phagocytes do not include:
 - A. Monocytes
 - B. Küffer cells
 - C. Kidney mesangial cells
 - D. Lymph node medullary cells
 - E. Endothelial cells
2. A polymorphonuclear neutrophil:
 - A. Is a bone marrow stem
 - B. Is closely similar to a mast cell
 - C. Contains morphological cytoplasmic granules
 - D. Is not a professional phagocytic cell
 - E. Has granules which stain with eosin
3. Which of the following is not produced following activation of the NADPH oxidase microbicidal pathway?
 - A. O_2
 - B. O_2^-
 - C. H_2O_2
 - D. NO
 - E. OH
4. Acute inflammation characteristically involves:
 - A. Constriction of arterioles
 - B. Capillary endothelial cell enlargement
 - C. Influx of macrophages
 - D. Influx of mast cells
 - E. Influx of neutrophils
5. Interferons:
 - A. Are found only in mammalian species
 - B. Are divided into five main families
 - C. Induce enzyme synthesis in target cell
 - D. Only affect infected cells
 - E. Are specific for individual viruses
6. Polymorphonuclear neutrophils attack bacteria:
 - A. Exclusively by oxygen-dependent mechanisms
 - B. Exclusively by oxygen-independent mechanisms
 - C. By phagocytosis
 - D. By secreting complement
 - E. By secreting interferon
7. Clonal selection occurs when antigen is encountered by:
 - A. Neutrophils
 - B. Mast cells
 - C. T cells
 - D. Basophils
 - E. Eosinophils
8. Horizontal transfer of acquired immune responsiveness involves the transfer of:
 - A. Antibody
 - B. Complement
 - C. Phagocytes
 - D. Lymphocytes

9. T-cell surface receptors for antigen partly recognize:
 A. Cytokines
 B. MHC
 C. ADCC
10. The basic Ig unit is composed of:
 A. 2 identical heavy and 2 identical light chains
 B. 2 identical heavy and 2 different light chains
 C. 2 different heavy and 2 identical light chains
 D. 2 different heavy and 2 different light chains
 E. Non-covalently bound polypeptide chains
11. The complementarity determinings:
 A. Are restricted to light chains
 B. Are in the constant part of the Ig molecule
 C. Binding to Fc receptors
 D. Are concerned in antigen recognition
 E. Occur at the C-terminal end of the Ig peptide
12. Ig idiotypes are found:
 A. In the constant region of the heavy chain
 B. In the constant region of the light chain
 C. In the hinge region
 D. In the variable region of both heavy and light
 E. Only in the light
13. A chromosome on which T-cell receptor and T-cell receptor alpha chain gene rearrangement has occurred lacks which of the following gene segments:
 A. Joining (J)
 B. Diversity (D)
 C. Variable (V)
 D. Constant (C)
 E. TCR beta chain
14. Somatic hypermutation of V genes
 A. Is confined to the hypervariable regions
 B. Involves heavy but not light chain immunoglobulin V gene
 C. Does not occur in IgM antibodies
 D. Can decrease the affinity of an antibody
 E. Is commonly seen for both antibody and TCR
15. The term variolation refers to:
 A. The generation of antibody variable regions
 B. The attenuation of virulent organisms
 C. Innoculation of scab material into small skin wounds
 D. The removal of scab material from an individual with smallpox
 E. A type of gene therapy
16. When a resting naïve T-cell engages its specific MHC/peptide complex displayed on the surface of a fibroblast it:
 A. Undergoes blast cell formation
 B. Produces IL-2
 C. Moves from G0 to G1 of the cell cycle
 D. Becomes anergic
 E. Secretes IL-1
17. Proliferation of activated T-cells:
 A. T-cell receptor alpha genes
 B. T-cell receptor beta genes
 C. T-cell receptor gamma genes
 D. T-cell receptor delta genes
 E. T-cell receptor gene receptor gene D region
18. Human monoclonal antibodies can be obtained:
 A. Using Epstein-Barr Virus immortalization of T-cells
 B. By human hybridomas selected for antibody production
 C. Using transgenic xenomouse strains
 D. By a single point mutation of a mouse monoclonal antibody

Bone marrow
Liver
Spleen
Yolk sac
Thymus

Stem cell factor is produced mainly by:

- A. Stem cells
- B. Thymic epithelium
- C. Bone marrow stromal cells
- D. Megakaryocytes
- E. The reticuloendothelial system

21. Which of the following statements might not be true?

- A. Pluripotent hematopoietic stem cells are produced in the bone marrow
- B. In adults, the thymus is virtually atrophied and is therefore no longer functional
- C. T-lymphocytes are educated in the thymus
- D. Not all lymphocytes are leukocytes
- E. Plasma cells are always derived from B-lymphocytes

22. Allelic exclusion is not seen for:

- A. T-cell receptor alpha genes
- B. T-cell receptor beta genes
- C. T-cell receptor gamma genes
- D. T-cell receptor delta genes
- E. T-cell receptor gene D region segments

23. When antigen reaches a lymph node in a primed animal:

- A. There is an increase in the output of cells in the efferent lymphatics over the following 24 hours
- B. There is a decrease in the output of cells in the efferent lymphatics over the following 24 hours
- C. There is an immediate output of activated blast cells
- D. It is transported to the spleen
- E. It is all immediately destroyed by macrophages

24. Which of the following lymphoid tissues is unencapsulated:

- A. Thymus
- B. Lymph node
- C. Spleen
- D. Tonsil
- E. MALT

25. RAG-1 and RAG-2 enzymes effect the recombination of:

- A. VDJ to CH
- B. H to L
- C. CDR1 to CDR2
- D. V to D
- E. VJ to CA

26. With reference to the variable Ig domain, which of these answers is false?

- A. It mediates the secondary consequences of antigen recognition
- B. It has parallel beta-plated sheet structures
- C. It uses beta-turn loops to bind antigen
- D. It has an extra long beta-turn relative to constant region domains
- E. It has a typical Ig fold with an intra-chain disulfide bond

27. Several of the complement components are:

- A. Glycolipids
- B. Cytokines
- C. Enzymes
- D. Hormones
- E. Antibodies

28. Which of the following gene clusters do not contribute to antigen binding:

- A. M
- B. CL
- C. CH
- D. D
- E. J

29. Expression of the CD3 antigen on a homodimer is characteristic of:

- A. Intraepithelial lymphocytes
- B. Follicular dendrite cells

- D. B-lymphocytes
 - E. Classical cytotoxic T-lymphocytes
 - F. Classical helper T-helper Tlymphocytes
30. The specialised cell type involved in the entry of lymphocytes into lymph nodes are
- A. M-cells
 - B. Mesangial cells
 - C. PALS
 - D. HEV endothelial cells
 - E. Selectin

31. Primary lymphoid organs include

- A. Thymus and spleen
- ☒ B. Thymus and bone marrow
- C. Thymus, bone marrow and spleen
- D. Thymus, bone marrow, spleen and lymph nodes

32. Which of the following is the site of T cell maturation?

- A. Bone marrow
- ☒ B. Thymus
- C. Spleen
- D. Appendix

33. All the following comes under non-specific defence mechanism except

- A. fever
- B. phagocytes
- ☒ C. cell mediated immunity
- D. complement system

34. Skin is the largest organ in the body and protects us by preventing pathogen entry. This is an example of

- ☒ A. Naturally acquired innate immunity
- B. Artificially acquired adaptive immunity
- C. Naturally acquired passive immunity
- D. Artificially acquired passive immunity

35. Which of the following substances will not stimulate an immune response unless they are bound to a larger molecule?

- ☒ A. Antigen
- B. Virus
- C. hapten
- D. MHC
- E. Antibody

36. Naturally acquired active immunity would be most likely acquired through which of the following processes?

- A. vaccination
- B. Drinking colostrum
- C. Natural birth
- D. infection with disease causing organism followed by recovery.

37. Cell mediated immunity is carried out by while humoral immunity is mainly carried out by

- A. B cells/T cells
- B. Epitopes/Antigens
- ☒ C. T cells/B cells
- D. Antibodies/Antigens
- E. Antibodies/Phagocytes

38. B cells that produce and release large amounts of antibody are called

- A. Memory cells
- B. Basophils
- ☒ C. Plasma cells
- D. Killer cells
- E. Neutrophils

39. System of over 20 blood proteins that form an enzymatic cascade to attack extracellular forms of infectious agents

... of the ...
... system ...
... image

- Which one of the following anatomical regions is most likely to show a predominance of T lymphocytes?
- A. A posterior iliac crest in the spleen
 - B. A Peyer's patch in the small intestine
 - C. A tonsillar follicle
 - D. The bone marrow
 - E. The germinal center of a lymph node follicle

Section B (60 marks)- answer all questions

- 41 a) How do phagocytes recognize pathogens? (5Mrks)
b) How does normal microbiota contribute to the innate immunity in the first line of defense? (5Mrks)
c) Natural Killer (NK) cells and Killer (K) cells provides innate and adaptive protections respectively; discuss? (10 marks)
- 42 a) Define the following terms and give one example in each case:
(i) Antigenic sin (2marks)
(ii) Cross reactivity (3marks)
(iii) Sero conversion (2marks)
(iv) Class switching (2 marks)
(b) Explain the immunological changes in the following phases of infections?
(i) Acute phase
(ii) Convalescent phase (4marks)
(c) With the aid of illustration (graph) describe the kinetics of primary and secondary immune response (7 Marks)
43. Provide the definitions for the following
(i) Primary lymphoid organs (2 marks)
(ii) Secondary lymphoid organs (2 marks)
b) Summarise the structure and physiological function of the following lymphoid organs
(i) Thymus (3 marks)
(ii) Lymphnode (3 marks)

END